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A review of wind turbine bearing condition monitoring: State of the art and challenges

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ABSTRACT

Since the early 1980s, wind power technology has experienced an immense growth with respect to both the turbine size and market share. As the demand for large-scale wind turbines and lor operation & maintenance cost continues to raise, the interest on condition monitoring system has increased rapidly. The main components of wind turbines are the focus of all CMS since they frequently cause high repair costs and equipment downtime. However, vast quantities of their failures are caused due to a bearing failure. Therefore, bearing condition monitoring becomes crucial. This paper aims at providing a state-of-the-art review on wind turbine bearing condition monitoring techniques such as acoustic measurement, electrical effects monitoring, power quality, temperature monitoring, wear debris analysis and vibration analysis. Furthermore, this paper will present a literature review and discuss several technical, financial and operational challenges from the purchase of the CMS to the wind farm monitoring stage.

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1. Introduction

The market forecasts that by 2018 the wind energy cumulative gigawatts (GW) will be 43% higher than of 2015's GW, as illustrated in Fig. 1 [1]. Nowadays, wind energy is a growing and reliable renewable energy source in the world. However, the industry still experiences premature main components (MC) failures, which increase the operation & maintenance (O&M) costs [2]. For instance, early cracks in bearings have been detected with less than 3 years of operation or 5–10% of the expected life [3]. Beyond that, it is estimated that the rate of gearbox failure is one incident per 145 turbines per year [4]. As the demand for wind energy continues to grow substantially, reducing O&M costs and improving reliability have become top priorities in wind turbine (WT) maintenance strategies. O&M cost is a priority because it is responsible for 20-30% of the life cycle cost on onshore installations and up to 30% on offshore installations [5]. In addition, reliability is a fundamental attribute that need to be guaranteed [6]. In the past decade the key research collaborations have focused on improving WT reliability, prediction and diagnosing incipient failures of WT sub-assemblies [7]. Manufactures are trying to improve the reliability of their systems, e.g. gearbox, through same configuration or by developing new configurations [8]. Furthermore, bearings cause high impact on cost and reliability. For instance, it has been shown that bearings cause more than 50% of faults on gearboxes, and therefore 50% of all costs associated to these faults [4,9]. Consequently, wind turbine bearing condition monitoring (WTBCM) has also become a priority to limit premature failures.

WTs are exposed to extremes, variable and around the clock 24/7 weather conditions, which generate rapid changes on temperature, air pressure, wind shear, wind speed and total load. Due to these variations, WTs undergo constantly changing global and local dynamics and loads. Therefore, the components have to work on intense and variable mechanical stress, which may lead to the occurrence of failures. Besides mechanical stress, the bearing temperature and the different temperatures cause temperature stress along the machine, e.g. shaft's and gears' temperature, together with lubrication problems can also accelerate bearings faults [10]. Also, for bearings at different locations the loads and stresses are different. For instance, misalignment in the drive train leads to abnormal loads and accelerates the wear of the bearings placed at that specific location [11]. To avoid such premature failures, it is necessary to develop better WT designs and also apply reliable and cost-effective condition-monitoring (CM) techniques [12]. Due to all these factors, condition monitoring systems (CMS) are expected to grow from 25 to 36.1% in 2015 according to sellers and buyers perspectives respectively [13].

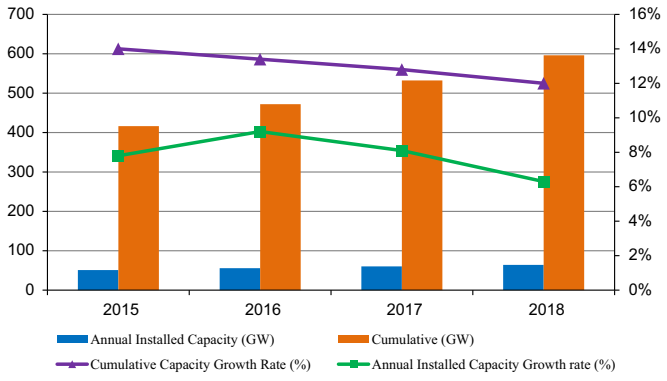


Fig. 1. Market forecast of wind energy for 2015–2018. Source: Adapted by the authors using [1].

This paper is structured in seven sections. Section 2 describes CM and why it is important to monitor bearings. Section 3 presents the state-of-the-art review on data acquisition techniques for condition monitoring applied to wind turbine's bearings. Section 4 emphasizes a literature review on fault detection and diagnoses methods used by different authors. Section 5 presents a literature review on remaining useful life techniques for bearings. Section 6 focuses on technical, financial and operational challenges to implement the CM. Finally, Section 7 presents the concluding remarks.

2. Condition monitoring definition and importance of bearings

First and foremost, it is important to have a clear understanding of what is CM and how to define it. Although each author presents CM a little bit differently, in general, one can say that CM is a monitoring process or a sensitive tool that focuses on early detection of faults, failures and wear of machinery with the intention to minimize downtimes and O&M costs, and consequently, maximize production [14–17]. Beyond that, since the CMS detects failure at an incipient phase it also optimizes maintenance planning and action, by reducing the chances of chain damage and catastrophic failures. In addition, the O&M allows understanding better what is happening to the machine, thus improving maintenance actions and practices and, consequently, operation [2] (Table 1).

CMS can also be defined, in a more general way, as a process or tool that integrates technologies, specialized people, condition indicators and quality data/ measurements to estimate the health condition of a machine and to make the best decisions about maintenance actions [15]. Depending on the technique the CMS is also able to give a probabilistic forecast of the future condition or the remaining useful life of the equipment (Table 2).

Secondly, it is essential to realize the impact of bearing's failures on wind turbines. Several condition monitoring papers focus on gearbox and generator studies because they cause high equipment downtime and therefore high loss of production and also because they have a high percentage of replacement over the wind farm's (WF) life [18]. However, recent studies show that bearings cause around 70% of gearbox downtime and 21–70% of generator downtime (21% on small generators ($P < 1$ MW), 70% on medium generators ($1\ MW < P < 2\ MW$) and 50% on large generators ($P > 2MW$)). Fig. 2 illustrates the downtime for gearboxes and medium generators [9,19]. In addition, corrective or preventive maintenance are not efficient enough to avoid these failures rates. Corrective maintenance is only performed after the failure has occurred, therefore it would not be able to avoid any failures. Preventive maintenance is a time-based maintenance that faces two main challenges: under-maintenance and over-maintenance. In order words, the system performance is not well monitored, resulting in unexpected failures or there are an excessive number of maintenance activities resulting in a waste of resources [20]. Therefore, it is crucial to use condition-based maintenance on bearings and also to develop tools and methods that can be used to early diagnose faults (Table 3).

3. Data Acquisition Techniques

3.1. Acoustic Measurement

Acoustic monitoring is similar to vibration monitoring; however, there are some main differences. Vibration sensors are rigid mounted to the component involved, while acoustic sensors are

Table 1
Summary table for Acoustic Measurement.

Reference number	Author(s), Year	Major contributions
[43]	Oh et al., 2012	Fan bearings wear detection by AE and evaluation of precursor parameters
[44]	Elforjani et al., 2010	AE for fault detection on slow speed shafts and bearings
[45]	Eftekharnejad et al., 2010	Bearing fault detection by spectral kurtosis, AE and vibration analysis
[46]	Kilundu et al., 2011	AE through cyclostationarity, spectral correlation and integrated indicators for bearing fault detection
[47]	Elforjani et al., 2012	AE to detect natural fault initiation and propagation on bearings
[48]	Nienhaus et al., 2012	High frequency AE burst-detection to identify faults on low speed roller bearings
[49]	Sandoval et al., 2013	AE method by tracking time-based fault indicators for bearing monitoring
[25]	Niknam et al., 2013	AE method to detect faults on bearings subject to unbalance
[50]	Jian et al., 2013	AE monitoring, cepstrum and wavelet analysis to identify bearing faults for linear motion stage.
[51]	Towsyfyan et al., 2013	AE time and frequency analysis on journal bearings under different conditions / with load and speed correlations.
[52]	Jia et al., 2013	Early subsurface cracks detection by AE and vibration analysis during fatigue test on bearings.
[53]	Law et al., 2013	Wavelet, Hilbert-Huang transform and multi-step screening processes to monitor a spindle bearing system via AE.
[54]	Ali et al., 2014	AE to detect and describe defects on bearings, including rate of degradation
[55]	Cárcel et al., 2014	AE on time and frequency domain with to spectral kurtosis (SK) as denoising tool to detect faults on bearings
[56]	Chacon et al., 2015	Combines Wavelet packet, AE signal denoising, Hilbert Transform for envelope extraction and autocorrelation to find bearing incipient defects.

Table 2
Summary table for Electrical Effects Monitoring.

Reference number	Author(s), Year	Major contributions
[57]	Renaudin et al., 2010	Fault detection based on instantaneous angular speed fluctuations through magnetic encoders applied to bearings
[58]	Holweger et al., 2012	Barkhausen noise measurement for fault detection on bearings
[59]	Neti et al., 2012	Electrical multi-phase imbalance separation technique to detect faults on bearings and gearbox
[60]	Whittle et al., 2013	Currents and electric stress analysis for fault detection on generators bearings
[61]	Harlişca et al., 2013	Stray magnetic flux monitoring proposed a statistical analysis of rolling bearings
[62]	Amirat et al., 2013	Homopolar current of the generator though an ensemble empirical mode decomposition used for bearing faults diagnosis
[63]	Oberg et al., 2013	Cracks detection based on gold tracks and electrical circuits
[64]	Romanenko et al., 2014	Electric discharge machining and post processing for fault detection on bearings
[65]	Machado et al., 2014	Electrical resistance measurement to diagnosis faults on bearings

attached to the component by flexible glue with low attenuation. Vibration sensors register the local motion, whereas, the acoustic sensors "listen" to the component through sound level meters [21,22]. These devices have a microphone that converts pressure levels and variations both internally and externally into a voltage signal, which is then recorded on a meter [23]. An acquisition device that has antialiasing and both a high sampling rate and dynamic range is ideal for this type of measurement [23,24] (Table 4).

The sources of sounds emitted from operating wind turbines can be divided into two categories: mechanical sounds and aerodynamic sounds, from the interaction of turbine components, and produced by the flow of air over the blades, respectively. In addition, there are four different types of sounds that can be classified from wind turbine noise readings: tonal, broadband, low frequency and impulsive which depends on noise characteristics [23]. From the noise readings, one can determine the higher-frequency components to predict possible faults. Furthermore, it is possible to validate the noise compliance of the turbine by measuring signals like the sound power level and by sending the acoustic data through third-octave analysis [24] (Table 5).

The most effective acoustic-based bearing health monitoring is acoustic emission (AE). It is a transient impulse caused by a rapid release of strain energy in solid material under stress conditions such as mechanical or thermal loadings. The main application of AE is the detection of cracks. Therefore, this technique can be used as a tool for condition monitoring of bearing faults and shaft cracks. Typically, the accuracy of these methods depends on sound pressure and sound intensity data [14,25,27].

The major advantage of this technology is its ability and sensitivity to capture surface and subsurface micro-damage. It has the potential of being more cost effective, simpler and more reliable

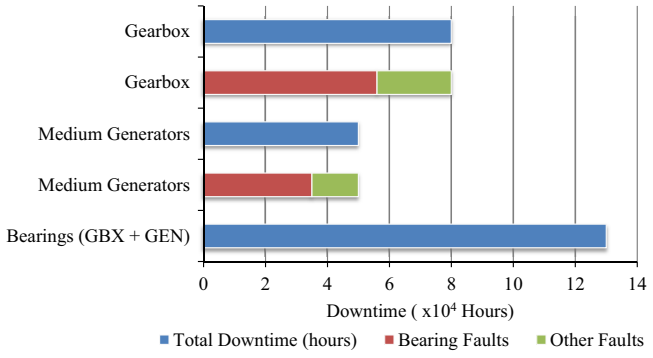


Fig. 2. Aggregated Downtime for Gearbox and Medium Generators (1 MW < P < 2 MW). Source: Adapted by the authors using [9] and [19].

CM technique [27]. So far, it has been proven that in some cases AE can ensure earlier fault detection than vibration-based condition monitoring systems [25].

3.2. Electrical effects monitoring

CM of electrical equipment such as motors, generators and accumulators is typically performed using voltage and current analysis looking for an unusual phenomenon [14]. For accumulators the impedance can be measured to establish the condition and capacity [21,22]. Electrical resistance can also be used for the structural evaluation of certain WT components since it varies with stiffness. Therefore, it is a method able to detect changes that can be caused by cracks, delamination, and fatigue [14]. This technology could be used to detect generators' bearings cracks and

Table 3
Summary table for Power Quality and Temperature Monitoring.

Reference number	Author(s), Year	Major contributions
[66]	Uluyol et al., 2011	Bearing's temperature obtained on SCADA is used for fault detection
[67]	Kim et al., 2012	Fault detection algorithm based on neural network used on bearings' temperature
[68]	Guo et al., 2012	NSET method for fault detection applied on generator bearings' temperature
[10]	Kusiak et al., 2012	NSET method for fault detection applied on generator bearings' temperature
[69]	Yang et al., 2013	Bearing fault detection through power and temperature variations
[70]	Astolfi et al., 2014	Temperature analysis for each power output on wind turbine subsytems, including bearings
[71]	Shahriar et al., 2015	Method based on power/speed of the WT applied to gearbox. Bearing fault is under study

Table 4
Summary table for Oil Debris Monitoring (ODM).

Reference number	Author(s), Year	Major contributions
[18]	Dupuis, 2010	Diagnostics and prognostics of bearing and gear failures modes
[72]	Bai et al., 2011	Faults detection through ferrography and spectrometric on large slewing bearings
[73]	Jiang et al., 2012	Post failure bearing analysis and how faults would be detected and failure avoided through oil samples
[74]	Jiang et al., 2012	Physical and chemicals analysis on gear oil samples and bearings faults are identified

Table 5
Summary table for Vibration Analysis (VA).

Reference number	Author, Year	Major contributions
[75]	Miao et al., 2011	Zoom interpolated discrete Fourier transform based on multiple modulations to identify faults
[76]	Jayaswal et al., 2011	Vibration analyses to early fault detection on bearings
[77]	Abdussiam et al., 2011	Bearings monitoring based on TESPAR (Time Encoded Signal Processing and Recognition), vibration and envelope analysis
[78]	Ziani et al., 2012	Bearing fault detection by Artificial Neural Network and Genetic Algorithm
[79]	Waters et al., 2013	Real time vibrations-based method to detect, localize, and identify a faulty bearing
[80]	Liu et al., 2013	Frequency domain, percent power, peak rms, sequential forward search algorithm and adaptive neuro fuzzy inference systems to detect and identify faults on bearings
[81]	Sarvajith et al., 2013	Fourier and Discrete Wavelet transform to determine bearings condition
[82]	Roulias et al., 2013	Wavelet denoising with neighblock threshold technique for condition monitoring of roller bearings
[83]	Borghesani et al., 2013	Squared envelope spectrum and the kurtosis of the corresponding band-pass filtered analytic signal for bearings defect diagnosis
[84]	Nizwan et al., 2013	Vibrational analysis and discrete wavelet transform for bearing fault detection using wavelet decomposition
[85]	Kumar, 2013	Discrete wavelet transform and artificial neural network to detect bearings faults
[86]	Sun et al., 2013	Discrete wavelet transform and envelope analysis to diagnose rolling bearing faults
[87]	Li et al., 2013	Morlet wavelets filter and spectral kurtosis to detect bearings defects
[88]	Zhao et al., 2013	Tacholess envelope order analysis technique to diagnose bearings faults
[89]	Sun et al., 2013	Multiwavelet denoising technique with a data-driven block threshold to detect rolling bearing defects
[90]	Liang et al., 2014	Intelligent bearing fault detection method based on a calculus enhanced energy operator to extract the bearing fault signatures
[91]	Unala et al., 2014	Fault estimation algorithm based on artificial neural network (ANN), envelope analysis, Hilbert Transform and Fast Fourier Transform to detect bearing faults
[92]	Dybala et al., 2014	Empirical Mode Decomposition to decompose the raw vibration signal into a number of Intrinsic Mode Functions to diagnose bearings' faults
[93]	Dalvand et al., 2014	Instantaneous frequency method with envelope analysis and condition indicators to detect bearing defects
[94]	Ma et al., 2014	Combined KICA and LS-SVM to achieve fault monitoring and classification of bearings
[95]	Shakya et al., 2014	Vibration data in time, frequency and time-frequency domain were analyzed using Mahalanobis Distance by application of Gram-Schmidt orthogonalization process and Chebyshev's inequality
[96]	Ahn et al., 2014	Fault detection of a roller bearing system using a wavelet denoising scheme and proper orthogonal value of an intrinsic mode function covariance matrix, which is obtained through empirical mode decomposition
[97]	Zimroz et al., 2014	Diagnostic approach for bearings using parameters such as peak-to-peak, root mean square of vibration acceleration and generator power
[98]	Wang et al., 2014	Four steps method based on signal processing to directly extract signal components of a rolling element bearing
[99]	Khanam et al., 2014	Discrete wavelet transform assisted by sym5 wavelet to detect and estimate defect size on bearings
[100]	Saidi et al., 2014	Combines the higher order spectra analysis features and support vector machine to identify different fault patterns of bearings
[101]	Zarei et al., 2014	Time-domain vibration signal and artificial neural networks to detect and classify bearing faults
[102]	Liu et al., 2014	Local mean decomposition technology and multi-scale entropy to diagnose roller bearing faults
[103]	Guo et al., 2014	Envelope extraction and independent component analysis for faults detection on rolling element bearing
[104]	Ali et al., 2015	Empirical mode decomposition, energy entropy, intrinsic mode functions and artificial neural network to identify and classify bearings defects
[105]	Caesarendra et al., 2015	Largest Lyapunov exponent algorithm for faults detection and deterioration track of low speed slew bearing
[106]	Ming et al., 2015	Cancellation method through iterative calculation of the envelope of the multi-component signal to detect bearings faults
[107]	Shakya et al., 2015	Integrated Mahalanobis–Taguchi–Gram–Schmidt method to fuse damage identification parameters to detect and classify bearings' defects

general faults. It is widely used for CM of rotating electric machine but has not been used in WT due to lack of experience [28].

Several techniques are available: discharge measurements; velocity measurements; contact force measurements and oil analysis. Basically, these techniques are used to detect medium / high voltage grid, switches, transformers and cabling isolation faults. Moreover, these types of inspection measurements do not directly influence the operation of the WT [14,21,22].

3.3. Power quality

Wind farms will be required to meet more and more grid compliance criteria since the energy matrix includes a greater participation of wind energy, which will continue to increase. In other words, wind power quality has to be better and more reliable [19]. In addition, both mechanical power measured on the drive shaft and electrical power measured on the generator are disturbed by mechanical and/or electrical faults. Therefore, some faults, such as bearings faults, can be detected depending on their influence on the mechanical and/or electrical energy. Furthermore, if a fault causes torsional changes in the equipment it can be detected through the mechanical power [12].

Power quality can be obtained through supervisory control and data acquisition (SCADA) systems. Statistical indicators can be used to indicate problems or simple flags on bearings that require maintenance actions or that can lead to an O&M improve. In addition, measurements can be made on the turbine's transformers to acquire the necessary data for the required analysis. In this case, clamps and sensors with high voltage and current inputs as well as isolation are ideal for these measurements [22,24].

3.4. Temperature Monitoring (TM)

TM of the WT components is one of the most common methods of CM. It can be used for preventive or predictive maintenance and it can be measured with a variety of different sensors. The temperature measurements of individual components such as bearings' temperature are very important since they bring more information and can be used in different types of analysis. It is important to highlight that some internal and external environment/machine conditions can influence temperature measurements. For instance, temperature varies with load, so by analyzing the evolution of temperature in a bearing it is important to know whether the temperature increased because of a fault or due to higher load. An effective and simple analysis to reduce the effect of load on the TM is to monitor the difference between temperatures, such as generator's not drive end bearing temperature minus generator's drive end bearing temperature. Since both temperatures would increase with load, their difference should be less dependent or practically immune to load variation.

There are several types of sensors that can be used for temperature measurement. However, the most commons are resistance temperature detectors (RTDs), optical pyrometers and resistant thermometers [12,24]. Every component and subcomponent has a temperature operational range; therefore if the TM is higher than the threshold it is possible to extract the information and detect a fault. Usually, the faults are caused by the deterioration of components or subcomponents from mechanical frictions or bad electrical connection [12,29]. In bearings and gears, frictions are usually caused by insufficient or inefficient lubricant properties and impacts due to misalignment that causes an extra heat and an increase in temperature, which can be detected using simple methods [25,26]. However, the disadvantage of TM is that it develops slowly and sometimes too late when compared to other CM methods. In other words, one could say that, alone, TM is not as efficient as other methods for early and precise fault detection [12,18].

3.5. Oil Debris Monitoring (ODM)

Wind turbine bearings and gears suffer continuous damage that causes splinters/debris. As the damage increases there will be more and more debris that are carried on the lubrication until end up in the oil filter [30]. In this method, the main goal is to detect the presence of metallic debris in the oil. This can be done through an online system, with sensitive sensors/particle counters installed on the wind turbine, or offline by taking periodic samples. In the online system, every time a particle passes through the sensor, an electrical pulse is generated. Knowing that each component/subcomponent has a different kind of particle with different properties and characteristics, one can determine the quality of the oil and therefore the existence of faults and wear [24,26,31]. In the offline system, one takes periodic samples and submit to oil analysis / spectrographic. As result, the concentration and characterization of the oil particles works as an indicator of faults and location [25,26].

Online ODM is a practical condition monitoring technique for the early detection and the tracking of damage. However, oil monitoring is mostly executed offline by taking periodic samples. On the other hand, this situation is changing and online oil monitoring is becoming more usual since it provides a direct measure of severity and progression of a fault just by the particle counting (without complex data analysis) [14,30,31].

3.6. Vibration Analysis (VA)

VA is the most common CM technology used in the industry for any kind of rotating equipment and it is an effective tool for the bearing fault diagnosis [14,32,33]. A vibration-based method allows preprocessing and post processing of the vibration data on time domain, frequency domain and time-frequency domain. Therefore, it is a versatile and complete tool to extract and detect useful information caused by most possible faults, wear and different types of damage either localized or distributed [26,33].

The vibration sensors used on WT are basically accelerometers. The sensor characteristics, the pre and post-processing, and the analysis methods for tower, main bearing and planet gears are different from the high speed drive train and generator due to low and high frequency ranges, respectively. In addition, these different frequency ranges add more challenges to wind turbine monitoring, mainly due to cross frequencies and low frequency components, which are harder to monitor. Both time and frequency domain signals can be used to diagnose faults. However, it is possible to extract more information from these signals through signal processing algorithms and alarms such as envelope signal and narrow band envelope alarms. Furthermore, there are several statistical condition indicators such as root mean square (RMS), crest factor, peak-to-peak, kurtosis and sidebands factors that can be used to extract useful information from measured vibration signals [34,35].

During the incipient phases of bearing damage there is a generation of high frequency signals, and also, the fault frequencies are close to the noise floor. Therefore, the vibration spectrum is less efficient at this phase of bearing fault. However, there is solid enveloping and demodulation methods applied on the high frequency content that clearly identify the bearing fault frequencies, and consequently, the location and type of fault [14,36,37].

Another way to monitor the vibration level is through wavelet transformations. It has been successfully applied to misalignment and bearing faults. In addition, it can be used as a general indication of a fault produced in a wind turbine [38]. Wavelet analysis

is based on a set of functions called daughter wavelets, which are obtained through time translation and amplitude dilation of a mother wavelet, able to map any signals in a way that is possible to have time-frequency representation of the signal. Therefore, it can be used to process non-stationary data and it is capable of quantifying/identifying the location and level of damage [39,40,41]. There are over 10 types of mother wavelets that can be used to do CM; however, to monitor bearings it is important to have mother wavelets that can adapt well to periodic impacts and modulations [42].

WA is a modern identification tool, which is still being evaluated and tested in different kinds of systems/environments. Its transforms have the feature to give better frequency resolution (but worse time resolution) at low frequencies and better time resolution (but worse frequency resolution) at high frequencies, which is usually necessary in wind turbine CM [41,42]. There are several papers that used wavelet transforms to detect and diagnose faults on WT bearings listed in the literature review. However, complex aliasing may exist in the high-frequency portion [14,40].

4. Review of fault detection and diagnosing methods for bearings

4.1. Acoustic Measurement

Oh et al. (2009) showed a study (including experiment simulating lubricant starvation) on the precursor parameters, such as acoustic noise, vibration and lubricant temperature to detect degradation on fan bearings. Acoustic emission was the best precursor parameter to represent the health of dry bearings [43].

Elforjani et al. (2010) applied AE monitoring on slow speed shafts and bearings. Experimental cases with different conditions were executed separately to shaft and bearings. Bearings' test demonstrated the efficiency of AE to detect crack initiation and propagation [44].

Eftekharnejad *et al.* (2010) used a test rig to simulate varying operating conditions for a bearing. Then, after comparison between vibration and AE measurements, it was shown that AE it more sensitive to detect incipient damage. The Spectral Kurtosis and kurtograms were also explored, and it was efficient in denoising both signals [45].

Kilundu et al. (2011) showed that the use of cyclostationarity for AE is very effective, mainly using spectral correlation and the integrated indicators for monitoring bearing outer race defect; however, it is not as effective with inner race defects [46].

Elforjani et al. (2012) presented a work to demonstrate the use of AE to detect natural defect initiation and propagation on rolling element bearing at a built test-rig [47].

Nienhaus et al. (2012) used high frequency AE to detect faults on low speed roller bearings. A test rig was mounted using an automated AE burst-detection to reduce amount of data. Comparing to vibration analysis, AE proved to be more efficient in terms of detecting faults earlier [48].

Sandoval et al. (2013) proposed an acoustic emission-based method for monitoring tapered roller bearings by tracking time-based fault indicators such as peak value and kurtosis. Experiments results highlighted the fault. However, the method was not able to distinguish inner from outer race faults [49].

Niknam et al. (2013) presented an AE based method for detecting a faulty bearing subject to unbalance. A multi-purpose test rig was used to analyze AE with various conditions and speed levels. PAC-energy was also analyzed and the results confirm the effectiveness of AE, although other sources of uncertainty (e.g stiffness) should be considered [25].

Jian et al. (2013) used AE monitoring, cepstrum and wavelet analysis to identify bearing defects for linear motion stage. Three frequency bands were analyzed and it is possible to diagnose the bearing condition [50].

Towsyfyan *et al.* (2013) investigate the use of AE on journal bearings under different conditions. A test rig was mounted to test two bearings supporting a motor, one at the not drive end and other at the drive end. Then time domain and frequency domain analysis were executed showing that AE signals were correlated to load, rotational speed and lubrication [51].

Jia *et al.* (2013) used AE to detect early subsurface cracks during a fatigue test on bearings. The bearing's out race was replaced to accelerate bearing wear and the test was conducted with AE and vibration sensors. The results show that AE was more efficient [52].

Law et al. (2013) applied a method combining wavelet, Hilbert-Huang transform and multi-step screening processes to monitor a spindle bearing system via AE. The method was tested and the AE signatures were correlated to the preload changes and temperature increase [53].

Ali et al. (2014) presented a study on the application of AE to monitor bearings health and more specifically, the ability to describe defect sizes and observe the rate of degradation [54].

Cárcel et al. (2014) used AE to monitor roller bearings focusing on spectral kurtosis (SK) as denoising tool. A test rig was mounted to test bearings in different conditions and the signals were analyzed on time and frequency domain. SK proved to be an efficient denoising tool and also improve the burst visibility [55].

Chacon et al. (2015) proposed a novel envelope analysis method for bearing incipient defect detection that is able to identify localized defects in an incipient stage, in which the signal-to-noise ratio is extremely low. This method combines Wavelet packet, for AE signal denoising, the Hilbert Transform for envelope extraction and autocorrelation function, to find patterns in the AE signal. Experimental results indicated that the proposed method was able to detect incipient defects with 9 dB lower SNR than traditional envelope analysis [56].

4.2. Electrical effects monitoring

Renaudin et al. (2010) proposed an alternative bearing condition monitoring based on instantaneous angular speed fluctuations, which are measured with magnetic encoders [57].

Holweger et al. (2012) applied Barkhausen noise measurement (also referred as Magnetoelastic or the Micromagnetic method) on a full bearing test rig. It has been showed that this method could lead to real predictive condition monitoring in highly dynamic loading environments [58].

Neti et al. (2012) suggested an electrical multi-phase imbalance separation technique (EMIST) to early detect faults mainly on bearings and gearbox of wind turbines. Experimental analyses were executed and they validated the method through generator's bearings inner race fault and gearbox pinion wear and tear [59].

Whittle et al. (2013) have investigated bearings failures due to currents and electric stress on wind turbine generators. Numerical simulations on the WT generator circuit show excess in electrostatic discharge machining of the bearing balls and race ways [60].

Harlişca et al. (2013) proposed a statistical analysis of rolling bearings based on stray magnetic flux monitoring. Experimental tests were conducted with different sensors and conditions. Also, three types of bearings faults were evaluated: outer race crack, outer race drilled hole and seal damage [61].

Amirat et al. (2013) monitored the homopolar current of the wind turbine generator though an ensemble empirical mode decomposition (EEMD) approach. The most energized modes of a bearing failure were identified and then analyzed on experimental data. It has been

showed that the 4th intrinsic mode function can be used as a condition indicator for bearing condition monitoring [62].

Oberg et al. (2013) developed a method to detect crack evolution during impacts based on gold tracks and electrical circuits. Experiments were conducted on steel ball bearings / grades of alumina to show the reliability of the method [63].

Romanenko et al. (2014) used different means to measure the electrical stress placed on the bearings, the resulting vibration signal, and apply signal processing for feature extraction through electric discharge machining [64].

Machado et al. (2014) used electrical resistance measurement and numerical experiments to diagnosis faults on bearings. This method is efficient to monitor bearing faults even comparing to vibration analysis [65].

4.3. Power Quality and Temperature Monitoring

Power quality and temperature monitoring were combined together because most bearings' fault detection and diagnosis methods use temperature measurements and power data obtained through SCADA.

Uluyol et al. (2011) used the power curve to assess the performance and components condition. It has showed that bearing SCADA data (mainly temperature) were used on condition based maintenance. An analytic power curve analysis separates prefailure from normal operation data [66].

Kim et al. (2012) studied fault detection systems based on wind vs. power performance curve obtained from SCADA. Generators and gearboxes' bearings temperature were also used in the analysis. A fault detection algorithm was developed using neural network and it was able to detect, effectively, the normal and abnormal behavior of wind turbine [67].

Guo et al. (2012) construct a method based on the Nonlinear State Estimate Technique (NSET) and applied to the generators bearing temperature. Simulations showed that the method can successfully detect early damage on bearings [68].

Kusiak et al. (2012) used data-mining approach based on Neural Networks algorithms to identify bearing faults through temperature changes on the wind turbine generator. Five overtemperature events were analyzed and predicted 1.5 h before fault occurrence [10].

Yang et al. (2013) used SCADA data to assess the health condition of the wind turbine systems. The technique was used to detect generator bearing fault through power and temperature variations. It has successfully detected the occurrence and the further development of the fault [69].

Astolfi et al. (2014) analyzed temperature data from the SCADA. Gearbox, generator and rotor bearings were examined on a real wind farm and several anomalous turbines were detected. The post processing is based on temperature analysis for each power output, and it has proven its efficiency [70].

Shahriar et al. (2015) proposed a novel condition monitoring method based on power/speed of the WT, although bearing fault is under study, its efficient has been proved for gearbox for varying wind speed [71].

4.4. Oil Debris Monitoring (ODM)

Dupuis (2010) presented a study on bearing and gear failures modes, diagnostics and prognostics. He also presents a technique through oil debris monitoring to count the cumulative number of particles and apply this technique to in-service wind turbines [18].

Bai et al. (2011) purposed a condition monitoring method based on oil analysis to detect faults on large slewing bearings. Ferrography and spectrometric analysis identified an abnormal bearing condition efficiently [72].

Jiang et al. (2012) presented a post failure analysis of a rolling bearing based on oil samples. It also shows that oil monitoring would be an effective condition monitoring method to avoid these failures and extend bearings' life [73].

Jiang et al. (2012) applied an oil monitoring technique based on physical and chemical analysis, spectroscopic, ferrographic and filtergram analysis of gear oil samples. The analysis indicated wear on the bearing and further quality checks are recommended [74].

4.5. Vibration Analysis (VA)

Miao et al. (2011) proposed a zoom interpolated discrete Fourier transform (zoom IpDFT) based on multiple modulations. It has been showed, through several experiments, that this method provides better performance especially on multi-fault situations with aliasing and multiple close frequencies. Also, the comparison of zoom IpDFT and FFT indicated that zoom IpDFT can identify faults accurately and with good resolution [75]

Jayaswal et al. (2011) presented vibration analyses techniques on bearings. They detect the earlier fault in bearing using vibration monitoring. By study the FFT spectrum of bearing vibration signal they access the condition of bearing [76].

Abdussiam et al. (2011) developed an approach using vibration signal to bearings monitoring based on TESPAR (Time Encoded Signal Processing and Recognition) and its combination with envelope analysis. As result, TESPAR gives reliable results only when combined with the envelope [77].

Ziani et al. (2012) compared the performance of bearing fault detection by using Artificial Neural Network (ANN) and Genetic Algorithm (GA). The time domain signals from normal and defective gear boxes are extracted and used as input to ANN. The system features are selected on the basis of genetic algorithm and linear Discriminant analyses (LDA) is used as evaluation function [78].

Waters et al. (2013) presented a real time vibrations-based method to detect, localize, and identify a faulty bearing in an ocean turbine electric motor [79].

Liu et al. (2013) measured radial acceleration of ball bearings and extracted peak amplitude of the frequency domain, percent power, and peak RMS. Then used Sequential Forward Search Algorithm and Adaptive Neuro Fuzzy Inference Systems to detect and identify the defects [80].

Sarvajith et al. (2013) used Fourier and Discrete Wavelet transform to determine rolling element condition in bearings. Preprocessed signals were then decomposed to several levels using discrete wavelet transform. Classification between healthy and faulty bearing and identification of fault was done using support vector machine. Results obtained were physically verified to ensure the effectiveness of the technique developed [81].

Roulias et al. (2013) applied wavelet denoising with NeighBlock threshold technique for condition monitoring of roller bearings. The condition monitoring efficiency of a statistical feature is assessed taking into account both raw and denoised bearing vibration signals. As a result, the technique showed that it was able to improve the estimation of the remaining useful life as well as the change point detection of the structural health of the asset [82].

Borghesani et al. (2013) showed analytical identities for the squared envelope spectrum and the kurtosis of the corresponding band-pass filtered analytic signal. Also, the analytical results showed a link to the cepstrum pre-whitening. Then they discussed about an optimal indicator for the choice of the demodulation band, the ratio of cyclic content (RCC), which endows the kurtosis with selectivity in the cyclic frequency domain and whose performance is compared with more traditional kurtosis-based indicators such as the protrugram [83].

Nizwan et al. (2013) presented a study of vibrational signal analysis for bearing fault detection using Discrete Wavelet Transform,

which was used to decompose the signal at different frequency scales. Experiments conducted with different types of speed and bearing defect indicated that wavelet decomposition analysis could be used effectively on bearing condition monitoring [84].

Kumar (2013) used Discrete Wavelet Transform to detect bearings faults on a test rig under one load and two speed conditions. The statistical features extracted from the dominant wavelet coefficients are used as inputs to an artificial neural network classifier to evaluate its performance. The results were compared between a raw and a denoised vibration signal [85].

Sun et al. (2013) combined discrete wavelet transform and envelope analysis to extract the characteristic spectrum of rolling bearing vibration data. The spectrum cross-correlation was applied to diagnose different faults on rolling bearings. The results showed the high accuracy and performance of the method [86].

Li et al. (2013) proposed an automatically frequency band selection method based on the Morlet wavelets filter and spectral kurtosis. The vibration signals of two bearings were analyzed using the proposed method. The results indicated that, comparing with the original envelope spectrum and the FIR filter based kurtogram, the proposed approach was effective and practical [87].

Zhao et al. (2013) proposed a tacholess envelope order analysis technique to diagnose bearings faults. First, a tacholess order tracking (TLOT) method was used to extract the tachometer information from the vibration signal itself. Then, an envelope order spectrum (EOS) was used to recover the bearing characteristic frequencies in the order domain. By combining the advantages of TLOT and EOS, the proposed technique was capable to detect effectively and accurately bearing faults under varying speeds, even without the use of a tachometer [88].

Sun et al. (2013) proposed a multiwavelet denoising technique with the data-driven block threshold. The data-driven block threshold selected the optimal block length and threshold at different decomposition levels by using the minimum Stein's unbiased risk estimate. The simulation experiment and the feature detection of a rolling bearing with a slight inner race defect indicated that the proposed method successfully detected the weak features of incipient faults [89].

Liang et al. (2014) proposed an intelligent bearing fault detection method based on a calculus enhanced energy operator to extract the bearing fault signatures in the presence of strong noise and multiple vibration interferences without prior information of the resonance excited by the bearing fault. This method does not require a bandpass filtering step and hence eliminates the burden to acquire the resonance information [90].

Unala et al. (2014) proposed a fault estimation algorithm based on artificial neural network (ANN) on extracted features using envelope analysis accompanied by Hilbert Transform and Fast Fourier Transform able to detect bearing faults. Then the ANN model was modified using a genetic algorithm providing, an optimal skillful fast-reacting network architecture with improved classification results [91].

Dybala et al. (2014) presented a method to diagnose rolling faults of bearings by using Empirical Mode Decomposition to decompose the raw vibration signal into a number of Intrinsic Mode Functions (IMFs). They used a new method of IMFs aggregation into three Combined Mode Functions (CMFs) to divide the vibration signal into three parts of signal: noise-only part, signal-only part and trend-only part. Experimental results on two different case studies showed that this method could diagnose incipient bearings' faults [92].

Dalvand et al. (2014) presented an instantaneous frequency method with envelope analysis of vibration signal and calculus of condition indicators such as kurtosis and crest factor to detect bearing defects. They also showed experimental results to prove the effectiveness of the method [93].

Ma et al. (2014) combined KICA and LS-SVM to achieve fault monitoring and classification of bearings. The vibration signal was mapped by kernel methods. When a fault was detected, the time domain and wavelet energy features were extracted to construct the multi-domain mixed feature set, and input the feature vector into LS-SVM for classification [94].

Shakya et al. (2014) proposed a methodology for online monitoring of rolling element bearing. Vibration data in time domain, frequency domain and time-frequency domain are used to identify critical parameters. The parameters were fused into Mahalanobis Distance by application of Gram-Schmidt orthogonalization process. Chebyshev's inequality was applied to the Mahalanobis Distance for online monitoring and damage stage detection [95].

Ahn et al. (2014) investigated fault detection of a roller bearing system using a wavelet denoising scheme and proper orthogonal value (POV) of an intrinsic mode function covariance matrix, which is obtained through empirical mode decomposition. It also showed that covariance matrices from healthy and damaged bearings exhibited different POV profiles, which could be a damage-sensitive feature [96].

Zimroz et al. (2014) proposed a diagnostic approach for bearings of wind turbine using parameters obtained from commercial diagnostic system such as peak-to-peak, root mean square of vibration acceleration and generator power that is related to the operating conditions. Then, they applied the method on two different bearings condition [97].

Wang et al. (2014) proposed a four steps method based on signal processing to directly extract signal components related to rotational speed from a time-frequency representation of a rolling element bearing. Experimental results showed the effectiveness of the method [98].

Khanam et al. (2014) presented a decomposition of the vibration signal by using discrete wavelet transform assisted by sym5 wavelet. The decomposed signal evidently splits the peak corresponding to the ball entry into and exit from the fault, enabling in an estimation of the defect size present in the bearing. Experiments conducted for different sizes of the defect present on the outer race of deep groove ball bearing affirmed the efficacy of the method with maximum deviation of 2.06% [99].

Saidi et al. (2014) presented a novel pattern classification approach for bearings diagnostics, which combines the higher order spectra analysis features and support vector machine classifier. Vibration bi-spectrum patterns were extracted and subjected to principal component analysis for dimensionality reduction. Then, the principal components were fed to support vector machine to distinguish bearing faults and levels of severity, which were measured in the experimental test bench running under different working conditions. The results indicated that the proposed method could reliably identify different fault patterns of rolling element bearings based on vibration signals [100].

Zarei et al. (2014) proposed a method to detect bearing defects of induction motors by submitting the time-domain vibration signal on two artificial neural networks, the first one to remove its non-bearing fault components, and the second one for fault classification through pattern recognition techniques. It was shown that better results were achieved when the filtered component of the vibration signal is used for fault classification rather than common methods that use directly vibration signal. Experimental results on three-phase induction motor verified the ability of the proposed method in fault diagnosis despite low quality (noisy) of measured vibration signal [101].

Liu et al. (2014) proposed a fault feature extraction method based on the local mean decomposition technology to decompose the non-stationary vibration signal of a roller bearing into a number of product functions and multi-scale entropy of each product function to be calculated as the feature vectors. The

analysis results from practical bearing vibration signals demonstrated that the proposed method was effective [102].

Guo et al. (2014) proposed a combination of envelope extraction and the independent component analysis (ICA) for feature extraction on rolling element bearing. The envelope allowed to obtain the impulsive response from faults, and also, to reduce the dimension of vibration sources in the sensor-picked signals. Then, the ICA algorithm was employed to separate the envelopes according to the independence of vibration sources. Experiments and simulations showed that the vibration features related to the REB faults could be effectively separated from disturbances and clearly exposed by the envelope spectrum [103].

Ali et al. (2015) proposed a feature extraction method based on empirical mode decomposition energy entropy of vibration signals. Then, they selected the most significant intrinsic mode functions to train an artificial neural network to classify bearings defects. Experimental results indicated a reliably tool to assess health condition and degradation of bearings [104].

Caesarendra et al. (2015) presented a new application of the largest Lyapunov exponent (LLE) algorithm for feature extraction on low speed slew bearing condition monitoring. Experimental results on lab and industrial slew bearings data indicated a better deterioration track than traditional vibration signal analysis [105].

Ming et al. (2015) proposed a novel deterministic component cancellation method based on the iterative calculation of the envelope of the multi-component signal (including both discrete frequency components and cyclic impulse responses), which is theoretically calculated by the Hilbert transform in both time and frequency domains. Then, simulations and experiments on bearings were used to validate the theoretical calculation and the proposed deterministic component cancellation method [106].

Shakya et al. (2015) developed a methodology for defect type identification in rolling element bearings using the integrated Mahalanobis–Taguchi–Gram–Schmidt (MTGS) method to fuse eleven damage identification parameters (DIP) from vibration data into a single parameter called Mahalanobis Distance (MD) and gain values. The gain value is used to identify the usefulness of DIP and the DIPs with positive gain are again fused into MD by using Gram–Schmidt Orthogonalization process (GSP) in order to calculate Gram–Schmidt Vectors (GSVs. They proposed a rank of the probability of existence of a type of bearing damage through Defect Occurrence Index to classify the defects [107].

5. Remaining useful life methods for bearings

Dempsey et al. (2011) used oil debris analysis to indicate bearing damage progression and estimate the RUL on roller bearings. Experiments were conduct at a test rig and results show that the oil debris can be used to indicate progression and RUL, but further tests are required due to variance in spall propagation rates [108].

Butler et al. (2012) presented a methodology for the estimation of the remaining useful life (RUL) of the main bearing for a commercial wind turbine. A residual model is used to highlight potentially faulty behavior, which is then post processed to provide a suitable signal for extrapolation using particle filters [109].

Butler et al. (2012) discussed failures of offshore wind turbines and purpose three prognostic methods to estimate the Remaining Useful life (RUL), hidden markov model, neural network, and particle filter [110].

Chen et al. (2012) purpose a prognostics model able to predict the RUL with lack of data. The model is based on multivariable support vector machine. Simulations and bearing run-to-failure experiments shows that bearing's remaining life is successfully predicted [111].

Soualhi (2014) presented an approach that combined the Hilbert-Huang transform, the support vector machine, and the support vector regression for the monitoring of ball bearings. This approach indicates health, classifies the degradation and estimates the remaining useful life. Experimental results indicated a consistent method to improve bearing condition monitoring [112].

Boškoski et al. (2015) proposed an approach to effectively estimate the RUL for bearing fault prognostics that employs Rényi entropy, Bayes' rule and non-parametric Gaussian process models. The proposed approach was evaluated on the dataset provided for the IEEE PHM 2012 Prognostic Data Challenge [113].

Ali et al. (2015) proposed a method based on the data-driven prognostic approach, and explored the combination of Simplified Fuzzy Adaptive Resonance Theory Map neural network and Weibull distribution to be able to evaluate the health state of bearings and to estimate their remaining useful life (RUL) by Prognostics and Health Management techniques. Experimental results showed that the proposed method could reliably predict the RUL of rolling element bearings based on vibration signals [114].

6. Technical, financial and operational challenges

There are several challenges to do WTBCM, however, we will focus on five main challenges that enclose technical, financial, operational and management issues from CMS purchase up to the WF monitoring stage. Some of these topics were discussed among the participants and during forums on the fourth annual wind farm data management and analysis on November 2014 [115].

I. The first of all challenges that a CMS faces is in the purchasing phase because the pay back of the system cannot be “exactly” measured upon contract. Also, the high cost (since a large WF require several CMS to equip the entire WF) and complexity of the CMS has discouraged WT operators [28]. When one purchases a CMS it is not possible to know upfront what faults the system will identify on the wind farm over the years and therefore how fast the investment will pay itself back. Consequently, WF owners have to rely on other WFs data and probability. To calculate financial gains of the system, after the system is installed, one estimates the financial gains of each fault detected/replaced due to the CMS. Therefore, over the years it is possible to calculate the pay back and the financial benefits of CMS. Pérez *et al.* applied a life cycle cost model on a real case study in Germany finding that the return of the investment will be after the 7th year of operation for an annual rate of return of 1 % [38].
II. Regarding to the monitoring techniques, another challenge is the hard integration between CMS techniques. For instance, some WF's operators use vibration based condition monitoring, oil debris and SCADA analysis, but it is not easy to integrate them in an automatic way so that the results are more reliable. In other words, there are challenges in conveying and interpreting datasets and results from such analyses in a convincing manner to WT specialists [116]. This is one of the reasons why the industry still experiences false alarms and misdiagnosis on its CMS.
III. CM methods should permit an easy automatization. Otherwise, these methods become not scalable, which means, once there are several WF and a large quantity of WT to monitor the system will not be able to handle it. As a result, automatization and scale ability turn into two essential attributes that all methods should have.
IV. Early prediction and replacement decision. WFs often have logistics issues for both onshore and offshore. As result, CM methods need to give a reasonable time before failure for

planning the corrective maintenance actions. For example, in Brazil several bearings are imported, depending on the case it may take months to arrive in the wind farm [117]. Also, some repairs need a crane that can also take a while to arrive in the WF. Therefore, it is indispensable that the CM method gives an early prediction. In addition to early warning, problem identification and continuous monitoring are also key characteristics of CM [118]. Once the problem is identified the system continuous to monitor until the time of replacement. It is also worthy to do visual inspections and comparison with other WTs.

V. How to set the alarms and the monitoring approach are other challenges because it is often necessary to have historical data from each WT. Each WT has a different behavior due to several reasons as cited on the introduction. Consequently, it is needed to measure each WT for few weeks/months to be able to set the alarms efficiently. In the future, it would be possible to perform O&M decisions based on past experience [116]. In addition, the monitoring approach could be improved, most of the CMS focus only on alarm management; which means, once the system indicates an alarm it checks the WT condition. However, if there are several WTs near but under the alarm level, the O&M will not be notified until the threshold is passed. It is necessary to also have an under standing of how well the whole WF is performing indepen dently of the alarms status of individual WTs.

7. Concluding remarks

In conclusion, it could be highlighted that wind turbine bearings should be monitored due to its high impact on downtime and component replacement. Most of gearbox and generators faults are due to bearings failures, and the bearing replacement cost is lower than the gearbox/ generator's replacement and downtime cost. Therefore, it is also very profitable to identify and replace faulty bearings.

WTBCM can be done through several commercial systems that use diverse techniques. However, it is possible to identify a significant trend in research opportunities in academic and industrial categories in the following areas:

I. Automatization, applicability and scalability of CMS
II. Methods and algorithms for earlier detection
III. Online CMS with smart diagnosis
IV. Estimation of the RUL

Many authors are looking for this system through pre and post processing and/or fusion of conventional techniques. Also, there are new methods being developed through different approaches of data interpretation and statistics.

The CMS challenges summarized some important aspects that sometimes are forgotten on condition monitoring. It is essential to always consider the financial ability and applicability of any CM method. We believe that there should be more papers commenting and describing these financial and practical challenges/aspects of their methodologies to make implementations on WFs easier and faster.

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